

OptiCrop: Smart Agricultural Production Optimization Engine

Sample Project Documentation

Project Description:

OptiCrop is an advanced Machine Learning-powered smart agricultural recommendation system designed to support modern, data-driven farming practices by providing accurate and intelligent crop prediction solutions. The system analyzes multiple critical environmental and soil-related parameters including nitrogen, phosphorus, and potassium (NPK) soil nutrient levels, temperature, humidity, soil pH, and rainfall patterns to determine the most suitable crop for cultivation under specific conditions.

Scenarios:

Scenario 1: Smart Crop Recommendation for Farmers

A farmer enters soil and environmental details such as N, P, K, temperature, humidity, pH level, and rainfall. The system analyzes the data and recommends the most suitable crop for maximum yield and better farming efficiency.

Scenario 2: Crop Suitability and Environmental Assessment

A user wants to understand whether current soil and climate conditions are suitable for a particular crop. The application evaluates input parameters and provides insights about crop compatibility and productivity potential.

Scenario 3: Agricultural Research and Policy Planning

An agricultural researcher or policymaker uses the system to analyze crop-environment relationships and identify patterns. The platform helps in making data-driven decisions for sustainable farming strategies and resource optimization.

Technical Architecture:

Layer	Component	Technology
Presentation Layer	Web Interface — Home, About, FindYourCrop pages	HTML5, CSS3, Bootstrap 5
Application Layer	Flask Backend — Routes, Input Handling, Prediction	Python 3.10+, Flask
ML Layer	Crop Prediction — Logistic Regression, K-Means	Scikit-learn, Pickle
Data Layer	Agricultural Dataset + Trained Model Storage	CSV (Pandas), model.pkl (Pickle)

Pre-requisites:

- **Python 3.10+:** <https://python.org>
- **Flask Framework:** <https://flask.palletsprojects.com>
- **Scikit-learn:** <https://scikit-learn.org/stable>
- **Pandas & NumPy:** <https://pandas.pydata.org>

- **Matplotlib & Seaborn:** <https://matplotlib.org>
- **Jupyter Notebook:** <https://jupyter.org>
- **Git Version Control:** <https://git-scm.com>

Project Workflow:

Milestone 1: Problem Understanding & Data Collection

- Define agricultural business problem and identify key soil/climate parameters
- Download Crop_recommendation.csv dataset from Kaggle (2200 records, 8 features)
- Import required Python libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

Milestone 2: Data Analysis & Preprocessing

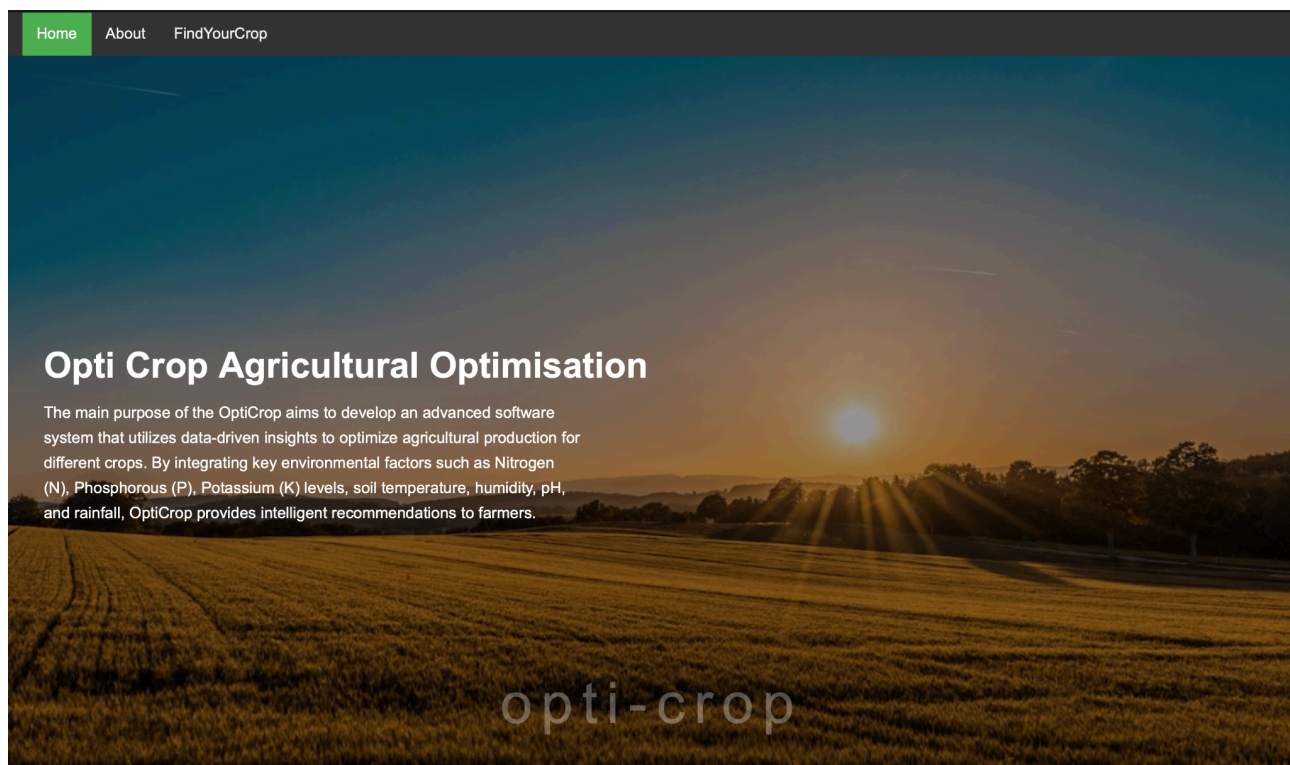
- Perform univariate, bivariate, and multivariate analysis on agricultural dataset
- Check for null values using `df.isnull().sum()` — dataset is clean with 0 null values
- Detect and handle outliers using IQR method and boxplot visualization
- Extract seasonal crops (Summer, Winter, Rainy) based on temperature and rainfall
- Split dataset into 80% training and 20% testing sets using `train_test_split()`

Milestone 3: Model Building

- Apply K-Means Clustering using Elbow method — optimal 4 clusters identified
- Train Logistic Regression model achieving 94% overall accuracy
- Evaluate model using classification report, confusion matrix, precision, recall, F1-score
- Save best model using `pickle.dump()` as `model.pkl` for deployment

Milestone 4: Flask Application Development

- Build `home.html`, `about.html`, `findyourcrop.html` using HTML5 and Bootstrap
- Develop `app.py` with Flask routes: `/`, `/about`, `/findyourcrop`, `/predict`
- Integrate trained `model.pkl` with Flask backend for real-time crop prediction
- Test application locally at `http://127.0.0.1:5000`



Milestone 5: Testing & Deployment

- Run complete application and test with multiple soil parameter inputs
- Verify prediction accuracy — input (N=105,P=35,K=40,T=25,H=64,pH=7,R=160) → coffee
- Push complete project to GitHub repository
- Record demo video and upload to Google Drive



Milestone 6: Conclusion

- OptiCrop successfully demonstrates intelligent crop recommendation using ML
- System achieves 94% prediction accuracy with Logistic Regression model
- Flask web application provides real-time accessible crop recommendations for farmers

Conclusion:

OptiCrop is an advanced Machine Learning-powered agricultural recommendation system that leverages multiple ML algorithms including K-Means Clustering and Logistic Regression to provide accurate and intelligent crop predictions. By combining agricultural science with artificial intelligence, the system analyzes N, P, K soil nutrients, temperature, humidity, pH, and rainfall patterns to recommend the most suitable crop for given conditions. The project achieves 94% prediction accuracy and successfully deploys as a Flask web application, making data-driven farming decisions accessible to farmers, researchers, and policymakers. The system reduces uncertainty in farming decisions and promotes efficient, sustainable agricultural production.